

AI Cost Summary for Paper Extraction

Claude · GPT · Gemini — cost per paper, based on our real papers (Edexcel AS Physics, ~30 question-paper pages + ~19 mark-scheme pages) · 9 October 2026

How AI pricing works (in short)

Input = what the AI reads

- Our instructions + each **page picture** of the question paper and mark scheme.
- A page picture counts as about **1,100–1,550 tokens** (fixed by its size).
- One paper ≈ **130,000–150,000 input tokens**. Input is the cheaper part.

Output = what the AI writes

- Questions, LaTeX maths, marks, topics, answers and diagram positions.
- Top models also "think" before answering; this is billed as output.
- One paper ≈ **13,000 tokens** written + **10,000–30,000** thinking. Output costs more per token.

Diagrams cost nothing extra. The AI must see each diagram, but it sees it **inside the page picture** it is already reading, and a page is priced by its size, not by what is drawn on it. So a page with a diagram costs the same as a page without one (already included in the prices below). The AI only writes the diagram's position (~40 tokens, under \$0.02 per paper), and the picture is **cut out on our own server for free**. The cost grows with the **number of pages**, not the number of diagrams.

Price list (per 1 million tokens)

Tier	Model	Input	Output	Use
Top	Claude Opus 5.5	\$4.00	\$20.00	best quality
Top	GPT-5.5	\$5.00	\$30.00	best GPT quality
Top	Gemini 3.1 Pro	\$2.00	\$12.00	best Gemini quality
Mid	Claude Sonnet 5.5	\$2.00	\$10.00	strong, half of Opus
Mid	GPT-5.4	\$2.50	\$15.00	strong, half of GPT-5.5
Mid	Gemini 3.5 Flash	\$1.50	\$9.00	fast, good value
Low	Claude Haiku 4.5	\$1.00	\$5.00	cheap, weaker on hard pages
Low	GPT-5.4 mini	\$0.75	\$4.50	cheap
Low	GPT-5 mini	\$0.25	\$2.00	very cheap, weakest
Low	Gemini 3.5 Flash-Lite	\$0.30	\$2.50	very cheap, weakest

Cost per paper (typical paper: ~30 QP pages + ~19 MS pages)

Tier	Model	Per paper	100 papers	1,000 papers	10,000 papers
Top	Claude Opus 5.5	\$1.05–1.45	\$105–145	\$1,050–1,450	\$10.5k–14.5k
Top	GPT-5.5	\$1.35–1.95	\$135–195	\$1,350–1,950	\$13.5k–19.5k
Top	Gemini 3.1 Pro	\$0.55–0.75	\$55–75	\$550–750	\$5.5k–7.5k
Mid	Claude Sonnet 5.5	\$0.55–0.75	\$55–75	\$550–750	\$5.5k–7.5k
Mid	GPT-5.4	\$0.65–0.95	\$65–95	\$650–950	\$6.5k–9.5k
Mid	Gemini 3.5 Flash	\$0.40–0.60	\$40–60	\$400–600	\$4k–6k
Low	Claude Haiku 4.5	~\$0.20–0.25	\$20–25	\$200–250	\$2k–2.5k
Low	GPT-5.4 mini	\$0.20–0.30	\$20–30	\$200–300	\$2k–3k
Low	GPT-5 mini	\$0.08–0.12	\$8–12	\$80–120	\$800–1.2k
Low	Gemini 3.5 Flash-Lite	\$0.10–0.15	\$10–15	\$100–150	\$1k–1.5k

Ranges depend on how much the model "thinks". Batch mode (results within hours) is about 50% cheaper on all three providers.

Effect of paper size (top-tier models, mid thinking level)

Paper size	Claude Opus 5.5	GPT-5.5	Gemini 3.1 Pro
Small: 20 QP pages + 10 MS pages	~\$0.80	~\$1.05	~\$0.42
Typical: 30 QP pages + 19 MS pages (our papers)	~\$1.25	~\$1.65	~\$0.65
Large: 40 QP pages + 30 MS pages	~\$1.85	~\$2.45	~\$1.00

The number of diagrams does not change these figures; more pages cost more because every page is read. Claude prices are Anthropic's official prices; GPT and Gemini prices are from recent public pricing pages (October 2026) and should be confirmed on the official OpenAI and Google pages.

What \$5 buys (pay-as-you-go API credit)

All three providers bill API use per token from prepaid credit or a monthly invoice. Chat subscriptions (Claude Pro, ChatGPT Plus, Gemini Advanced) do **not** include API use; the app needs an API key with credit. Claude's starting API tier opens with a first credit purchase of about \$5.

Tier	Model	Input tokens for \$5	Output tokens for \$5	Our papers for \$5
Top	Claude Opus 5.5	1.25 million	250,000	~3–4
Top	GPT-5.5	1.0 million	~167,000	~2–3
Top	Gemini 3.1 Pro	2.5 million	~417,000	~6–9
Mid	Claude Sonnet 5.5	2.5 million	500,000	~6–9
Mid	GPT-5.4	2.0 million	~333,000	~5–7
Mid	Gemini 3.5 Flash	~3.3 million	~556,000	~8–12
Low	Claude Haiku 4.5	5.0 million	1.0 million	~20–25
Low	GPT-5.4 mini	~6.7 million	~1.1 million	~16–25
Low	GPT-5 mini	20 million	2.5 million	~40–60
Low	Gemini 3.5 Flash-Lite	~16.7 million	2.0 million	~33–50

"Input tokens for \$5" = what \$5 buys if spent only on reading; "Output tokens" = only on writing. A paper uses both, so the last column is the real figure (from the cost-per-paper table above).

Our paper, worked through (typical Edexcel AS Physics paper: 30 question pages + 19 mark-scheme pages)

- **Requests:** ~38 (one per question page, ~10 for the mark scheme pages).
- **Input (what the AI reads):** ~140,000 tokens: every page picture (~1,100–1,550 tokens each) plus our instructions.
- **Output (what the AI writes):** ~13,000 tokens of questions, maths, marks, topics and answers, plus ~10,000–30,000 tokens of "thinking" on top models: ~23,000–43,000 in total.
- **With Claude Opus 5.5:** input $\approx 140,000 \times \$4 / 1\text{M} = \mathbf{\$0.56}$; output $\approx 23,000\text{--}43,000 \times \$20 / 1\text{M} = \mathbf{\$0.46\text{--}0.86}$; total $\approx \mathbf{\$1.05\text{--}1.45}$ per paper, so **\$5 reads about 3–4 papers**.
- **Same paper on Claude Sonnet 5.5:** $\approx \$0.55\text{--}0.75$, so \$5 reads about 6–9 papers. On Gemini 3.1 Pro: about the same. On GPT-5.5: $\approx \$1.35\text{--}1.95$, about 2–3 papers.
- **Worksheet crops and PDF downloads cost nothing:** the question and mark-scheme pictures are cut on our own server from the PDF's text. The AI is only asked where the parts are for papers without usable text.

Rate limits (how fast we can send work to the AI)

Every AI provider limits how much an account can send in a given time. Going over the limit returns a "too many requests, please wait" error. Limits are not a cost, but they decide **how long an upload takes**.

Limit	Meaning
Requests per minute (RPM)	how many AI calls the account can make each minute
Tokens per minute (TPM)	how many tokens (input and/or output) per minute
Requests / tokens per day	daily cap, mainly on free keys

Claude (Anthropic)

Tier	Requests / min	Input tokens / min	Output tokens / min	How you reach it
Starting (Tier 1 / Start)	~50	~50,000	~10,000	first credit purchase (~\$5)
Tier 2	~1,000	higher	higher	more spend
Tier 3	~2,000	higher	higher	more spend
Tier 4	~4,000	much higher	much higher	higher spend

Since June 2026 Anthropic groups accounts as Start → Build → Scale (older guides use Tier 1–4). New accounts may also have a daily spend cap (around \$100/day reported). Batch mode has its own, larger limits.

ChatGPT / OpenAI (example: GPT-5.5)

Tier	Requests / min	Tokens / min	How you reach it
Tier 1	500	500,000	first payment
Tier 2	5,000	1,000,000	\$50 total spent
Tier 3	5,000	2,000,000	\$100 spent
Tier 4	10,000	4,000,000	\$250 spent
Tier 5	15,000	40,000,000	\$1,000 spent + 30 days

Tiers rise automatically with spend. A single GPT-5.5 request over 272k tokens is billed at a higher rate; ours are about 3–5k tokens per request.

Gemini (Google)

Tier	Requests / min	Tokens / min	Requests / day
Free	~5–15 (by model)	~250,000	~100–1,000
Tier 1 (paid)	~150–300	~1,000,000	~1,500
Tier 2	~500–1,500	~2,000,000	~10,000

Pro models (e.g. Gemini 3.1 Pro) are paid-only since April 2026. Google sets exact limits per project and shows them in the AI Studio dashboard.

What the limits mean for our papers

One paper ≈ 38 requests, ~150,000 input tokens and ~35,000 output tokens.

Account	Time per paper (limit-wise)	Papers per day (roughly)
Claude starting tier	~3–4 min (token limits)	~15 per hour, no daily request cap
Claude Tier 2+	~1–2 min	hundreds
OpenAI Tier 1	~1–2 min (limits are high enough)	hundreds
Gemini free	~3–8 min	only ~2–26 per day (daily request cap)
Gemini Tier 1 (paid)	~1–2 min	~39 per day (1,500 requests/day cap)
Gemini Tier 2	~1 min	~260 per day

Our app reads 3 pages at a time, so on any paid tier a paper takes a few minutes in practice. Limit figures are from Anthropic's rate-limit docs and recent public guides (October 2026); they change and depend on the account, so confirm them in each provider's console.

For real use

- Use a **paid key**: higher limits, a paper in a few minutes, no daily stop.
- Use **batch mode** for bulk uploads: about half price and separate, larger limits.
- For hundreds of papers, raise the account tier first or spread uploads over time.

Students are not affected

- The AI is used **only when an admin uploads a paper** (once per paper).
- Browsing, search, similar questions and PDF downloads run on our own server and database: **no AI tokens and no AI limits**.
- A visitor limit on our own server can be added later (e.g. in nginx) to stop abuse.

Measured: our 10 papers read with Claude Code (Max plan) vs. the app with an API key

Our first 10 papers were read on 5 October 2026 by Claude Code (Claude Opus 5.5), one agent per paper. Claude Code logs the tokens of every step, so these input figures are **measured**. The written output is counted from the JSON each agent wrote. Thinking is hidden in the logs, so it is estimated at 5,000–20,000 tokens.

Paper	Steps	New input read	Input re-read (cached)	Output written	Same work at API price
Specimen 2015	20	109,484	1,195,931	~13,600	\$1.40–1.70
June 2016	16	99,894	831,638	~16,200	\$1.26–1.56
June 2017	7	141,197	462,495	~10,500	\$1.20–1.50
June 2018	15	83,481	728,543	~13,400	\$1.08–1.38
June 2019	7	131,349	433,282	~10,500	\$1.14–1.44
Oct 2020	9	146,870	747,909	~10,400	\$1.34–1.64
Oct 2021	14	90,974	727,873	~11,900	\$1.08–1.38
June 2022	7	155,890	494,493	~13,700	\$1.35–1.65
June 2023	7	149,209	485,567	~9,400	\$1.23–1.53
June 2024 (8PH0_s24_01)	7	159,566	513,432	~9,100	\$1.28–1.58
Average per paper	11	126,800	662,100	~11,900	\$1.24–1.54

API price used: Claude Opus 5.5 (input \$4 / 1M; new input written to the cache \$5 / 1M; cached re-reads \$0.40 / 1M, i.e. 10% of the input price; output \$20 / 1M). "Output written" is the extracted JSON (questions, LaTeX, marks, answers), about 3 characters per token. Hidden thinking is estimated and included in the price range. Not counted: the main Claude Code conversation that started the agents, and the one-time diagram-crop review (about \$0.80 per paper more at API price).

Claude Code on our Max plan

- One paper $\approx$ **790,000 input tokens** processed (127k new + 662k re-read) and $\sim$ 12,000 written, plus thinking.
- Claude Code re-sends the conversation at every step, so it moves **5–6 $\times$ more input** than the app. Most of it is cached re-reads at 10% of the price.
- **No money per paper:** on the Max plan this uses the plan's usage allowance (5-hour and weekly limits), not pay-per-token credit. **/usage** in Claude Code shows how much is left.
- Only for an admin doing the work by hand in Claude Code. The website cannot use the Max plan.

The app with an API key (pay per token)

- One paper $\approx$ **140,000 input + 23,000–43,000 output** tokens (13k written + 10k–30k thinking), in $\sim$ 38 requests.
- Claude Opus 5.5: input $\approx$ \$0.56 + output $\approx$ \$0.46–0.86 = **\$1.05–1.45 per paper**.
- Claude Sonnet 5.5: **\$0.55–0.75**. Batch mode: about half of either.
- This is what the website uses (uploads, re-process, Regenerate). Admin $\rightarrow$ Settings shows the requests made, and the console.anthropic.com Usage page shows the exact tokens and dollars.

In short: per paper, the money cost is about the same either way ($\sim$ \$1.25 at API price). With Claude Code on the Max plan it comes out of the subscription allowance; with the API key it is billed from credit.

Claude Max plan (~\$100 / month) vs. \$100 of API credit

Input tokens = what Claude reads

- Our instructions, page pictures, files and the conversation so far.
- **New input:** read for the first time (full price, or 1.25× when it is saved to the cache).
- **Cached input:** the same text read again on the next step, at **10% of the price**. Claude Code works this way all the time.

Output tokens = what Claude writes

- Answers, code, the extracted questions (JSON, LaTeX) and its hidden "thinking".
- Costs **5× more per token** than input (Opus 5.5: \$20 vs \$4 per million).
- 1 token $\approx \frac{3}{4}$ of an English word, or about 3–4 characters.

	Claude Max plan (Max 5×, ~\$100 / month)	API key with \$100 credit
How you pay	Fixed monthly fee	Pay per token from prepaid credit
Token limit	Anthropic publishes no token number . You get 5× the Pro plan's usage per 5-hour window , plus a weekly cap . Claude chat and Claude Code share the same allowance.	Exactly what the credit buys (table below). There is no time window: credit is used until it runs out.
When the limit is reached	You wait for the 5-hour window or the weekly reset (or add paid usage credits)	Requests fail until more credit is added
Who / what can use it	A person working in Claude Code or claude.ai (an admin doing work by hand)	Any software, including our website (uploads, re-process, Regenerate)
Can the website use it?	No: the plan is for personal use in Claude apps only	Yes: this is what the website needs
Unused allowance	Lost at each reset	Credit stays in the account until it is used
Exact usage shown in	claude.ai → Settings → Usage, or /usage in Claude Code (as a %)	console.anthropic.com → Usage (tokens and dollars)

What \$100 of API credit buys

Model	Only new input	Only cached input	Only output	Our papers (via the app)	With batch mode
Claude Opus 5.5 (\$4 / \$20)	25 million	250 million	5 million	~70–95	~140–190
Claude Sonnet 5.5 (\$2 / \$10)	50 million	500 million	10 million	~135–180	~270–360
Claude Haiku 4.5 (\$1 / \$5)	100 million	1,000 million	20 million	~400–500	~800–1,000

A real job always uses a mix of input and output, so the "papers" column is the useful one. It comes from the cost per paper above (Opus \$1.05–1.45, Sonnet \$0.55–0.75, Haiku \$0.20–0.25).

What our Max plan actually gave us (measured from Claude Code's logs, 3–9 October 2026)

Day	Claude replies	New input	Cached input	Output	Total tokens	Same at API price (Opus 5.5)
3 Oct	255	0.42 M	66.0 M	0.30 M	66.7 M	\$34
5 Oct (10 papers read)	497	4.30 M	180.2 M	0.29 M	184.8 M	\$99
6 Oct	756	1.88 M	249.7 M	0.58 M	252.2 M	\$121
7 Oct	400	1.09 M	170.8 M	0.30 M	172.2 M	\$80
8 Oct	396	2.21 M	182.3 M	0.37 M	184.9 M	\$91
9 Oct	319	1.10 M	183.5 M	0.29 M	184.9 M	\$85
Total, 6 working days	2,623	11.0 M	1,032.5 M	2.1 M	≈ 1,046 M	≈ \$510

This covers all the Claude Code work on this project: building the app as well as reading the papers. The output figures are a lower bound, because Claude Code's logs keep only part of each reply's output count, so the real API value is somewhat higher. 99% of the tokens are cached re-reads, which is why about 1 billion tokens comes to about \$510 and not thousands.

What this means

- In 6 days our ~\$100 Max plan did work worth **about \$510 at API prices**, about 5× the monthly fee, without paying per token.
- \$100 of API credit would have run out on about the **second day** of the same work.
- For hands-on work in Claude Code (building, fixing, reading papers by hand), the **Max plan is far better value**.

But the website still needs the API key

- The Max plan cannot be connected to the website. Uploads, re-process and Regenerate need an **API key**.
- \$100 of API credit ≈ **70–95 papers on Opus 5.5**, ~135–180 on Sonnet 5.5, about double with batch mode.
- Best of both: use **Max** for admin work in Claude Code, and keep a **small API credit** for the website's uploads.

Max limits: Anthropic publishes only multiples of the Pro plan (Max 5× ≈ \$100, Max 20× ≈ \$200 per month), a 5-hour window and a weekly cap. It does not publish token numbers, and the limits changed several times in 2026, so check claude.ai → Settings → Usage for the live figures.